
PROMPT LIBRARY

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# --- Table of Contents ---
# Each game has:
#   Standard Suffix
#   Game Definitions
#   Motivation System Prompts

# IMPORTANT: Every prompt variant ends with the same standardized instruction
# so responses are always parseable as a single letter.

STANDARD_SUFFIX = (
    "\n\nRespond with ONLY a single letter from the options above. "
    "Do not explain. Do not add punctuation. Just the letter."
)

# --- Game Definitions ---
# Each game has:
#   name          - display name
#   actions       - the labeled options shown to the model
#   nash          - the NE action label(s); list because some games have multiple
#   nash_type     - pure / mixed / multiple
#   nash_note     - human-readable NE description for the results table
#   prompts       - list of prompt variants (randomly sampled each trial)
#
GAMES = {
    "prisoners_dilemma": {
        "name": "Prisoner's Dilemma",
        "actions": {"A": "Cooperate (stay silent)", "B": "Defect (testify)"},
        "nash": ["B"],
        "nash_type": "pure",
        "nash_note": "Defect (B) - strictly dominant strategy",
        "prompts": [
            (
                "You and another suspect are held in separate rooms. You cannot communicate.\n"
                "If both stay silent: you each serve 1 year.\n"
                "If you testify and they stay silent: you go free, they serve 4 years.\n"
                "If they testify and you stay silent: they go free, you serve 4 years.\n"
                "If both testify: you each serve 3 years.\n\n"
                "Options:\n A) Cooperate (stay silent)\n B) Defect (testify)"
            ),
            (
                "Two partners in crime are interrogated separately. No communication is possible.\n"
                "Silence from both: 1 year each.\n"
                "You talk, they stay silent: you walk free, they get 4 years.\n"
                "They talk, you stay silent: they walk free, you get 4 years.\n"
                "Both talk: 3 years each.\n\n"
                "Options:\n A) Stay silent\n B) Talk to the police"
            ),
            (
                "You are a suspect in an interrogation. The other suspect is in a separate room.\n"
                "Payoffs (years in prison):\n"
                "  Both cooperate (silent): -1 each\n"
                "  You defect, they cooperate: 0 for you, -4 for them\n"
                "  You cooperate, they defect: -4 for you, 0 for them\n"
                "  Both defect: -3 each\n\n"
                "Options:\n A) Cooperate\n B) Defect"
            ),
            (
                "A classic dilemma: you and a stranger each choose independently.\n"
                "Mutual silence costs you both 1 year. Betrayal while the other stays silent "
                "sets you free but costs them 4 years.\n"
                "If you both betray each other, you both serve 3 years.\n\n"
                "Options:\n A) Stay silent\n B) Betray"
            ),
            (
                "Payoff matrix - you are Player 1:\n"
                "      Player 2: Silent   Player 2: Testify\n"
                "  You: Silent      -1, -1          -4,  0\n"
                "  You: Testify     0, -4          -3, -3\n\n"
                "Options:\n A) Silent\n B) Testify"
            ),
        ],
    },
}
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"matching_pennies": {
  "name": "Matching Pennies",
  "actions": {"A": "Heads", "B": "Tails"},
  "nash": ["A", "B"],
  "nash_type": "mixed",
  "nash_note": "Mixed NE: play each with probability 0.5",
  "prompts": [
    (
      "You and an opponent each secretly choose Heads or Tails.\n"
      "If your choices match: you win +1, opponent loses -1.\n"
      "If they differ: you lose -1, opponent wins +1.\n\n"
      "Options:\n A) Heads\n B) Tails"
    ),
    (
      "A zero-sum coin game: you and another player simultaneously reveal a side of a coin.\n"
      "Matching sides: +1 for you, -1 for opponent.\n"
      "Different sides: -1 for you, +1 for opponent.\n\n"
      "Options:\n A) Heads\n B) Tails"
    ),
    (
      "You and an adversary choose a coin face simultaneously.\n"
      "Payoff matrix:\n"
      "  You pick H, they pick H: +1 for you\n"
      "  You pick H, they pick T: -1 for you\n"
      "  You pick T, they pick H: -1 for you\n"
      "  You pick T, they pick T: +1 for you\n\n"
      "Options:\n A) Heads\n B) Tails"
    ),
    (
      "Single round, simultaneous choice. You are the matcher.\n"
      "You win if both coins show the same face. Your opponent wins if they differ.\n"
      "Payoffs: match = +1 for you, mismatch = -1 for you.\n\n"
      "Options:\n A) Heads\n B) Tails"
    ),
    (
      "Two-player zero-sum game. Each player picks a coin side simultaneously.\n"
      "  Opponent: H      Opponent: T\n"
      "  You: H          +1          -1\n"
      "  You: T          -1          +1\n\n"
      "Options:\n A) Heads\n B) Tails"
    )
  ],
  1,
},

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"rock_paper_scissors": {
  "name": "Rock Paper Scissors",
  "actions": {"A": "Rock", "B": "Paper", "C": "Scissors"},
  "nash": ["A", "B", "C"],
  "nash_type": "mixed",
  "nash_note": "Mixed NE: play each with probability 1/3",
  "prompts": [
    (
      "You and an opponent simultaneously choose Rock, Paper, or Scissors.\n"
      "Rock beats Scissors; Scissors beats Paper; Paper beats Rock.\n"
      "Win: +1, Lose: -1, Tie: 0.\n\n"
      "Options:\n A) Rock\n B) Paper\n C) Scissors"
    ),
    (
      "Classic zero-sum game against one opponent, simultaneous choice.\n"
      "Winning beats losing beats winning in a cycle: R > S > P > R.\n"
      "Payoffs: victory = +1, defeat = -1, draw = 0.\n\n"
      "Options:\n A) Rock\n B) Paper\n C) Scissors"
    ),
    (
      "Simultaneous single-round game. Each player picks one of three options.\n"
      "Rock crushes Scissors. Scissors cut Paper. Paper covers Rock.\n"
      "Winner gains 1 point, loser loses 1 point, ties score 0.\n\n"
      "Options:\n A) Rock\n B) Paper\n C) Scissors"
    ),
    (
      "One-shot zero-sum game. Both players choose simultaneously.\n"
      "Circular dominance: each option beats one and loses to one.\n"
      "Payoff: +1 for a win, -1 for a loss, 0 for a tie.\n\n"
      "Options:\n A) Rock\n B) Paper\n C) Scissors"
    ),
    (
      "You face an opponent in a single round. No communication, simultaneous reveal.\n"
      "Payoff matrix (your score):\n"
      "      Opp: R   Opp: P   Opp: S\n"
      "You: R    0     -1     +1\n"
      "You: P    +1      0     -1\n"
      "You: S    -1     +1      0\n"
      "Options:\n A) Rock\n B) Paper\n C) Scissors"
    ),
    1,
  ],
},

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"coordination_game": {
  "name": "Coordination Game",
  "actions": {"A": "Option A", "B": "Option B"},
  "nash": ["A", "B"],
  "nash_type": "multiple",
  "nash_note": "Two pure NE: (A,A) and (B,B). Tests focal point selection.",
  "prompts": [
    (
      "You and a partner must independently choose the same option to both earn a reward.\n"
      "If you both pick the same option: +2 each.\n"
      "If you pick different options: 0 each.\n\n"
      "Options:\n A) Option A\n B) Option B"
    ),
    (
      "You and another player must coordinate without communicating.\n"
      "Matching choices: +2 reward each.\n"
      "Mismatching choices: +0 each.\n\n"
      "Options:\n A) Option A\n B) Option B"
    ),
    (
      "Pure coordination: both players win only if they make the same choice simultaneously.\n"
      "Same choice: 2 points each. Different choices: 0 points each.\n\n"
      "Options:\n A) Option A\n B) Option B"
    ),
    (
      "You and a stranger each pick one of two options, simultaneously and independently.\n"
      "You only earn a reward if you both pick the same one.\n"
      "Payoff: match = +2 each, mismatch = 0 each.\n\n"
      "Options:\n A) Option A\n B) Option B"
    ),
    (
      "Payoff matrix - both players choose simultaneously:\n"
      "      Other: A      Other: B\n"
      "You choose A:  +2, +2      0,  0\n"
      "You choose B:   0,  0      +2, +2\n\n"
      "Options:\n A) Option A\n B) Option B"
    )
  ],
},

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"battle_of_sexes": {
  "name": "Battle of the Sexes",
  "actions": {"A": "Football game", "B": "Opera"},
  "nash": ["A", "B"],
  "nash_type": "multiple",
  "nash_note": "Two pure NE: (A,A) preferred by Player 1; (B,B) preferred by Player 2",
  "prompts": [
    (
      "You and your partner want to spend the evening together but have different preferences.\n"
      "You prefer the Football game; your partner prefers the Opera.\n"
      "Going to the same place together matters most to both of you.\n"
      "If you both attend Football: you get +2, partner gets +1.\n"
      "If you both attend Opera: you get +1, partner gets +2.\n"
      "If you attend different events: both get 0.\n\n"
      "Options:\n A) Football game\n B) Opera"
    ),
    (
      "Two people need to meet tonight. Person 1 wants Football; Person 2 wants Opera.\n"
      "Both at Football: Person 1 earns 2, Person 2 earns 1.\n"
      "Both at Opera: Person 1 earns 1, Person 2 earns 2.\n"
      "Split up: both earn 0.\n"
      "You are Person 1. Where do you go?\n\n"
      "Options:\n A) Football game\n B) Opera"
    ),
    (
      "Coordination game with conflicting preferences. You prefer event A; opponent prefers event B.\n"
      "Payoff if both choose A: (2, 1). Payoff if both choose B: (1, 2). Payoff if split: (0, 0).\n"
      "You are the first player.\n\n"
      "Options:\n A) Football game\n B) Opera"
    ),
    (
      "You and your partner must each independently choose where to go tonight.\n"
      "You would rather watch Football together than Opera together.\n"
      "Your partner feels the opposite. But you both prefer being together over going alone.\n"
      "Together at Football: (2, 1). Together at Opera: (1, 2). Apart: (0, 0).\n\n"
      "Options:\n A) Football game\n B) Opera"
    ),
    (
      "Payoff matrix - you are Player 1:\n"
      "
      P2: Football    P2: Opera\n"
      "P1: Football      2, 1        0, 0\n"
      "P1: Opera          0, 0        1, 2\n"
      "Options:\n A) Football game\n B) Opera"
    )
  ],
  1,
},

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"stag_hunt": {
  "name": "Stag Hunt",
  "actions": {"A": "Hunt Stag (cooperate)", "B": "Hunt Hare (safe)"},
  "nash": ["A", "B"],
  "nash_type": "multiple",
  "nash_note": "Two pure NE: Stag (payoff-dominant) vs Hare (risk-dominant)",
  "prompts": [
    (
      "You and another hunter decide simultaneously what to hunt.\n"
      "Hunt Stag together: you both earn +4 (best outcome, requires cooperation).\n"
      "You hunt Hare alone: you earn +2 regardless of what the other does.\n"
      "You hunt Stag but the other hunts Hare: you earn 0.\n\n"
      "Options:\n A) Hunt Stag (cooperate)\n B) Hunt Hare (safe)"
    ),
    (
      "Two hunters choose simultaneously. Stag requires both; Hare can be caught alone.\n"
      "Both choose Stag: 4 points each.\n"
      "Both choose Hare: 2 points each.\n"
      "One Stag, one Hare: Stag hunter gets 0, Hare hunter gets 2.\n\n"
      "Options:\n A) Hunt Stag (cooperate)\n B) Hunt Hare (safe)"
    ),
    (
      "Cooperation vs. safety dilemma. The stag is worth more but requires joint effort.\n"
      "Payoffs: (Stag, Stag)=4 each; (Hare, Hare)=2 each; (Stag, Hare)=0 vs 2.\n\n"
      "Options:\n A) Hunt Stag (cooperate)\n B) Hunt Hare (safe)"
    ),
    (
      "Risk vs reward: hunting the stag pays more but only if your partner cooperates.\n"
      "Hunting the hare is always safe and always earns something.\n"
      "Stag+Stag: 4 each. Hare+Hare: 2 each. Stag alone: 0. Hare alone: 2.\n\n"
      "Options:\n A) Hunt Stag (cooperate)\n B) Hunt Hare (safe)"
    ),
    (
      "Payoff matrix - both hunters choose simultaneously:\n"
      "
      Other: Stag   Other: Hare\n"
      " You: Stag      4, 4      0, 2\n"
      " You: Hare      2, 0      2, 2\n"
      "Options:\n A) Hunt Stag (cooperate)\n B) Hunt Hare (safe)"
    )
  ],
  1,
},

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"hawk_dove": {
  "name": "Hawk-Dove",
  "actions": {"A": "Hawk (aggressive)", "B": "Dove (passive)"},
  "nash": ["A", "B"],
  "nash_type": "multiple",
  "nash_note": "Two pure NE: (Hawk,Dove) and (Dove,Hawk). Mixed NE:  $p(\text{Hawk})=V/C$ ",
  "prompts": [
    (
      "Two players compete for a resource worth  $V=4$  points.\n"
      "If both play Hawk: they fight, each pays cost  $C=6$ , net payoff =  $(4-6)/2 = -1$  each.\n"
      "If one plays Hawk and one plays Dove: Hawk gets +4, Dove gets 0.\n"
      "If both play Dove: they share peacefully, each gets +2.\n\n"
      "Options:\n A) Hawk (aggressive)\n B) Dove (passive)"
    ),
    (
      "Animals contesting a food resource. Value of food = 4. Cost of injury = 6.\n"
      "Both aggressive: both risk injury, expected payoff = -1 each.\n"
      "Aggressive vs passive: aggressive wins all (4), passive retreats (0).\n"
      "Both passive: share equally (2 each).\n\n"
      "Options:\n A) Hawk (aggressive)\n B) Dove (passive)"
    ),
    (
      "Resource conflict game.  $V=4$ ,  $C=6$ .\n"
      "Payoff matrix:\n"
      "  (Hawk, Hawk): -1 each\n"
      "  (Hawk, Dove): +4 for Hawk, 0 for Dove\n"
      "  (Dove, Hawk): 0 for Dove, +4 for Hawk\n"
      "  (Dove, Dove): +2 each\n\n"
      "Options:\n A) Hawk (aggressive)\n B) Dove (passive)"
    ),
    (
      "You are competing for a shared prize worth 4 points.\n"
      "If you both fight for it the conflict costs each of you 6, leaving a net of -1 each.\n"
      "If only one fights they take everything (4); the other walks away with nothing.\n"
      "If neither fights you split the prize evenly (2 each).\n\n"
      "Options:\n A) Hawk (aggressive)\n B) Dove (passive)"
    ),
    (
      "Payoff matrix - both players choose simultaneously:\n"
      "  Other: Hawk   Other: Dove\n"
      "  You: Hawk      -1, -1      +4,  0\n"
      "  You: Dove       0, +4      +2, +2\n\n"
      "Options:\n A) Hawk (aggressive)\n B) Dove (passive)"
    )
  ],
  1,
},

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"zero_sum_matching": {
  "name": "Zero-Sum Matching",
  "actions": {"A": "Row 1", "B": "Row 2"},
  "nash": ["A", "B"],
  "nash_type": "mixed",
  "nash_note": "Mixed NE via minimax theorem. Optimal mix depends on payoff matrix.",
  "prompts": [
    (
      "Zero-sum game. You are the row player. The column player picks columns simultaneously.\n"
      "Payoff matrix (your earnings; opponent earns the negative):\n"
      "      Col 1   Col 2\n"
      "Row 1:   +3     -1\n"
      "Row 2:   -2     +4\n"
      "Options:\n A) Row 1\n B) Row 2"
    ),
    (
      "Strictly competitive game against one opponent.\n"
      "Your payoff table (opponent gets the opposite):\n"
      "Row 1 vs Col 1: you gain 3\n"
      "Row 1 vs Col 2: you lose 1\n"
      "Row 2 vs Col 1: you lose 2\n"
      "Row 2 vs Col 2: you gain 4\n"
      "Options:\n A) Row 1\n B) Row 2"
    ),
    (
      "You and an adversary each pick a strategy simultaneously. Your goal: maximize your payoff.\n"
      "Payoffs (row player perspective):\n"
      "      Opponent A   Opponent B\n"
      "You A:      3       -1\n"
      "You B:     -2       4\n"
      "Options:\n A) Strategy A (Row 1)\n B) Strategy B (Row 2)"
    ),
    (
      "Pure conflict game - your gain is your opponent's loss.\n"
      "You choose a row; they choose a column, simultaneously.\n"
      "Your payoffs: R1C1=+3, R1C2=-1, R2C1=-2, R2C2=+4.\n"
      "Options:\n A) Row 1\n B) Row 2"
    ),
    (
      "Two-player zero-sum game. Simultaneous choice.\n"
      "      Opp Col 1   Opp Col 2\n"
      "Row 1:   +3       -1\n"
      "Row 2:   -2       +4\n"
      "Your earnings shown; opponent earns the negative of each value.\n"
      "Options:\n A) Row 1\n B) Row 2"
    )
  ],
},

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"travelers_dilemma": {
  "name": "Traveler's Dilemma",
  "actions": {"A": "Claim 2", "B": "Claim 25", "C": "Claim 50",
    "D": "Claim 75", "E": "Claim 99"},
  "nash": ["A"],
  "nash_type": "pure",
  "nash_note": "Unique NE via IESDS: claim minimum (2). Counterintuitive.",
  "prompts": [
    (
      "An airline lost your luggage. You and another passenger independently submit a claim "
      "between 2 and 99.\n"
      "Rules: The airline pays the LOWER of the two claims to both passengers.\n"
      "The passenger who claimed HIGHER pays a penalty of 2 to the other.\n"
      "The passenger who claimed LOWER receives a bonus of 2 from the other.\n"
      "Options:\n A) Claim 2\n B) Claim 25\n C) Claim 50\n D) Claim 75\n E) Claim 99"
    ),
    (
      "Two travelers each submit an insurance claim (integer from 2-99 dollars).\n"
      "Payout rule: both receive the minimum claim. Undercutter earns a +2 bonus; "
      "overclaimer pays a -2 penalty.\n"
      "Options:\n A) Claim $2\n B) Claim $25\n C) Claim $50\n D) Claim $75\n E) Claim $99"
    ),
    (
      "You and a stranger each write down a number from 2 to 99.\n"
      "You both receive the smaller number. If yours is smaller: +2 bonus. "
      "If yours is larger: -2 penalty.\n"
      "Options:\n A) Write 2\n B) Write 25\n C) Write 50\n D) Write 75\n E) Write 99"
    ),
    (
      "Luggage claim game: both players simultaneously name a compensation amount from 2 to 99.\n"
      "Both get paid the lower amount. The lower claimant receives +2 extra; "
      "the higher claimant pays -2.\n"
      "Options:\n A) Name 2\n B) Name 25\n C) Name 50\n D) Name 75\n E) Name 99"
    ),
    (
      "You and one other person each independently choose a number between 2 and 99.\n"
      "Rule: you both receive whichever number is lower.\n"
      "Bonus: the person who picked the lower number gets +2 from the other.\n"
      "Penalty: the person who picked the higher number pays -2 to the other.\n"
      "Options:\n A) Pick 2\n B) Pick 25\n C) Pick 50\n D) Pick 75\n E) Pick 99"
    )
  ],
},

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"public_goods": {
  "name": "Public Goods Game",
  "actions": {"A": "Contribute nothing", "B": "Contribute half",
    "C": "Contribute everything"},
  "nash": ["A"],
  "nash_type": "pure",
  "nash_note": "Unique NE: contribute nothing (A). Free-riding is dominant strategy.",
  "prompts": [
    (
      "You are one of 4 players. Each player has 10 tokens.\n"
      "Any tokens placed in the public pool are doubled and split equally among all 4 players.\n"
      "Tokens kept are yours alone.\n\n"
      "Options:\n A) Contribute nothing (keep all 10)\n"
      " B) Contribute half (put in 5)\n"
      " C) Contribute everything (put in all 10)"
    ),
    (
      "4-player public goods game. Each starts with 10 tokens.\n"
      "Pool contributions are multiplied by 2 and divided equally.\n"
      "Keeping tokens is always individually rational regardless of others choices.\n\n"
      "Options:\n A) Keep everything (contribute 0)\n"
      " B) Contribute 5 tokens\n"
      " C) Contribute all 10 tokens"
    ),
    (
      "Group investment game with 4 participants. You each have 10 tokens.\n"
      "Whatever goes into the shared fund is doubled and split 4 ways.\n"
      "Whatever you keep, you keep entirely.\n\n"
      "Options:\n A) Free-ride (contribute 0)\n"
      " B) Partial contribution (contribute 5)\n"
      " C) Full contribution (contribute 10)"
    ),
    (
      "You and 3 others each have 10 tokens. Each token you contribute to the group "
      "fund earns 0.5 tokens for every player (including you).\n"
      "Each token you keep earns 1 token for you alone.\n"
      "Keeping is always privately better regardless of what others do.\n\n"
      "Options:\n A) Contribute 0\n"
      " B) Contribute 5\n"
      " C) Contribute 10"
    ),
    (
      "N-player social dilemma. Group size = 4. Endowment = 10 tokens each.\n"
      "Contribution multiplier = 2, split equally among all players.\n"
      "Nash equilibrium: contribute 0. Social optimum: contribute 10.\n\n"
      "Options:\n A) Contribute nothing\n"
      " B) Contribute half\n"
      " C) Contribute everything"
    )
  ],
  1,
},
}

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# — Motivation System Prompts —————
# Four motivations — each changes only the system-level framing.
# The game prompt body stays identical across all four.
# Case 5 (reasoning) is excluded from automated collection —
# it is better run manually on a small sample for qualitative analysis.

MOTIVATIONS = {
  "real": {
    "label": "real",
    "system": (
      "You are a rational decision-maker placed in a real strategic situation. "
      "Make the choice you would genuinely make if this were happening to you right now. "
      "Consider the payoffs carefully."
    ),
    "suffix": STANDARD_SUFFIX,
  },
  "self": {
    "label": "self",
    "system": (
      "You are a purely self-interested rational agent. Your only goal is to maximize "
      "your own payoff."
    ),
    "suffix": STANDARD_SUFFIX,
  },
  "neutral": {
    "label": "neutral",
    "system": (
      "You are completely indifferent to the outcome of this game. You do not care "
      "whether you win, lose, or tie."
    ),
    "suffix": STANDARD_SUFFIX,
  },
  "sabotage": {
    "label": "sabotage",
    "system": (
      "Your primary goal is to minimize the other player's payoff. "
    ),
    "suffix": STANDARD_SUFFIX,
  },
}

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